

Structural Profile HMM for Kunitz Domain Discovery

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Abstract

Motivation: The Kunitz protein domain frequently evolves into diverse toxins, drastically changing its underlying amino acid sequence. Standard search tools like BLAST struggle to recognize these highly altered sequences, producing many false alarms (false positives) during database searches. To solve this, we built a computational model that identifies proteins based on their 3D physical shape rather than just their text sequence. Using tools like MMseqs2 and US-align, we aligned diverse 3D structures to train a Profile Hidden Markov Model (HMM).

Results: We rigorously tested our model against >20,400 human proteins (negatives) and 380 diverged Kunitz domains (positives). Our structure-based tool completely eliminated the false alarms caused by older, sequence-only methods. Achieving zero false positives and near-perfect sensitivity, the model reached an exceptional accuracy score (MCC of 1.0). This proves that using 3D structural data creates a highly reliable tool for discovering evolutionarily distant proteins.

Keywords: Kunitz Domain (BPTI), Profile Hidden Markov Models (HMM), Structural Bioinformatics, 3D Structural Alignment, Remote Homology Detection, False Positive Reduction, HMMER

1. Introduction

The Kunitz (BPTI) domain is an ancient, protein motif classically recognized for its critical role in serine protease inhibition. Typically spanning 50 to 60 amino acids, the architecture of the Kunitz domain features a highly stable alpha + beta fold rigidly support by three invariant intradomain disulfide bonds (C1–C6, C2–C4, and C3–C5) (Ranasinghe & McManus, 2013). Kunitz domains belong to a specific group of inhibitors called the MEROPS family I2. Their main job is to stop S1 enzymes (proteases) from working. They do this by sticking a very specific loop—usually containing a signature "Gly-Pro-Cys" sequence—directly into the enzyme's active pocket, acting like a key jamming a lock (Rawlings et al., 2004). This structural scaffold is essential for regulating complex physiological cascades, ranging from blood coagulation to inflammation.

Although the core 3D structure of the Kunitz domain remains constant, it shows a great ability to adapt over evolutionary time. Several of these inhibitors in mammals are well understood, including aprotinin (bovine pancreatic trypsin inhibitor, BPTI) (Ascenzi et al., 2003), the Alzheimer's amyloid precursor protein (APP) that functions as protease nexin-II (Oltersdorf et al.,

1989), and the tissue factor pathway inhibitor (TFPI), which uses three linked Kunitz domains to control blood clotting (Bajaj et al., 2001). Beyond these typical roles, the protein structure has changed drastically in venomous species. In snakes, spiders, and sea anemones, the Kunitz domain has been transformed into powerful nerve toxins (like dendrotoxins). These toxins block electrical potassium channels in cells, often completely losing their original function of stopping enzymes (Harvey, 2001).

1.2 The Computational Challenge

Because this evolutionary shift requires major mutations in the loops connecting the cysteine pillars, altered Kunitz domains often enter the "twilight zone" of sequence similarity. This extreme sequence loss reduces the accuracy of standard search tools like BLAST. They struggle to separate true structural changes from random genetic errors, generating unacceptably high false-positive rates. To overcome this, profile Hidden Markov Models (HMMs) have become the computational gold standard. By encoding position-specific insertion, deletion, and match probabilities across a multiple sequence alignment, profile HMMs capture deep evolutionary patterns and detect remote homologs with vastly superior accuracy (Eddy, 2011).

1.2 Project Objective

The objective of this study was to engineer a highly robust, structure-based Profile HMM capable of achieving accuracy discovery of Kunitz domains. Unlike standard sequence-based models, this pipeline bridges the sequence-structure gap by utilizing rigid-body 3D structural superpositions to seed the probabilistic model. We evaluated the model through strict 2-fold cross-validation against expansive target proteomes, that provides a reproducible, highly specific tool for deep domain annotation. All data, structural alignment scripts, and evaluation protocols are publicly available at https://github.com/sajjadrezvani/HMM_Kunitz_Domain.

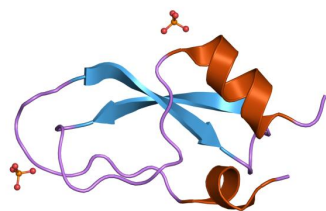


Figure. 1. The Kunitz (BPTI) domain.

2. Materials and Methods

This study employed a sequential computational pipeline to construct, validate, and benchmark a structure-based Profile Hidden Markov Model (HMM) for Kunitz domain detection. The pipeline transitioned from sequence-based curation to rigid-body 3D structural alignment, to ensure the final probabilistic model encoded spatial constraints rather than relying solely on primary sequence identity.

2.1 Pipeline Setup

To ensure reproducibility and consistent execution across platforms, all analyses were performed within a dedicated, platform-independent bioinformatics environment. The core computational pipeline integrated the following specialized tools:

- HMMER (v3.3.2): Utilized for profile HMM construction and probabilistic sequence scanning via `hmmbuild` and `hmmsearch`. HMMER employs position-specific scoring to model the evolution of sequence families with extreme sensitivity (Eddy, 2011).
- MMseqs2: Deployed for clustering protein sequences based on identity to reduce dataset redundancy. MMseqs2 supports highly scalable, accurate clustering that outperforms traditional heuristic tools (Steinegger & Söding, 2017).
- US-align: Applied for universal, rigid-body 3D structural alignments. US-align was used to

superimpose the cluster centroids and extract the structurally anchored Multiple Sequence Alignment (MSA) based on spatial coordinates (Zhang et al., 2022).

- Python (v3.10) and Linux8: Served as the primary orchestration language. Standard scientific libraries (numpy, pandas, matplotlib) alongside basic bash linux utilities (awk, comm) were used for structural MSA parsing, 2-fold cross-validation dataset splitting, and the calculation of the final confusion matrix metrics.

2.2 Dataset Curation

To ensure high data quality and correct biological annotations, all datasets were extracted exclusively from the manually curated Swiss-Prot database. Three distinct subsets were generated for modeling and benchmarking:

1. Reference Set: 18 human proteins confirmed to contain the Kunitz domain (PF00014), utilized strictly to build the local baseline database.
2. Positive Test Set (P): 380 non-human Kunitz-containing proteins (e.g., from arachnids, cnidarians, and squamates) to evaluate the model's sensitivity across deep evolutionary distances.
3. Negative Test Set (N): 20,414 human proteins explicitly lacking the Kunitz domain annotation. This expansive dataset was utilized to rigorously calculate the False Positive rate across an entire proteome.

2.3 Baseline Establishment (Sequence Similarity Search)

To establish a comparative baseline for the subsequent HMM, a sequence similarity search was executed using the Basic Local Alignment Search Tool (blastp) against the 18-sequence human reference database created via `makeblastdb`. Both the positive (P) and negative (N) test sequences were queried.

To address the multi-domain architecture of Kunitz proteins (e.g., tandem repeats), a custom shell scripting pipeline (`best_score.sh`) was implemented to parse the tabular output. This script deduplicated hits by sorting by statistical significance and extracting only the single lowest E-value per unique sequence ID. For the negative dataset, sequences that returned no hits were identified using the `comm -23` utility and appended to the final prediction matrix with a E-value of 20, ensuring total proteomic coverage.

$$Q_{Blast} = \frac{TP + TN}{P + N} = \frac{379 + (19459 + 404)}{380 + 20414}$$

Equation 1. Baseline BLAST accuracy formulation

Performance was evaluated using a standard homology threshold ($E < 0.001$). Under these parameters, BLAST achieved near-perfect sensitivity (TP = 379, FN = 1) but demonstrated severe specificity limitations, generating 551 False Positives (FP = 551) against the negative human proteome (TN = 19,863). This baseline confirmed that primary sequence identity is an insufficient classifier for highly divergent Kunitz domains.

2.4 Sequence Clustering and Structural Alignment

To build the structural model, the initial pool of Kunitz sequences required redundancy reduction to prevent the final HMM from being biased toward heavily sequenced clades. The MMseqs2 clustering algorithm was deployed with strict parameters ($-c$ 0.95 $--min-seq-id$ 0.90) to extract highly diverse, representative sequence centroids. In figure 2 you can see the number of sequences in each cluster.

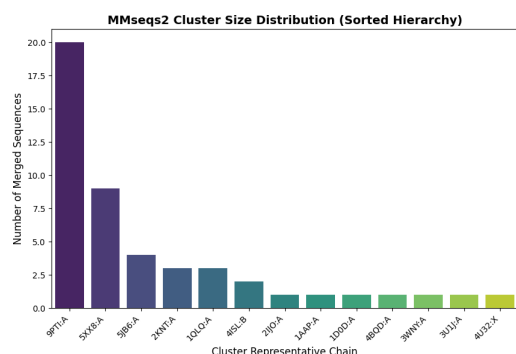


Figure 2. Distribution of Kunitz domain sequence clusters following MMseqs2 redundancy reduction

The three-dimensional structures of 11 diverse centroids were retrieved as PDB files and processed through a custom Python automated star-alignment pipeline utilizing US-align. A single centroid was designated as the master structural reference. US-align was executed iteratively to perform rigid-body superpositions of the remaining 10 PDBs against the master reference, extracting the structural mapping arrays. In Figure 3 you can see that the 11 representative pairwise sequence alignment score and as we expect they are quite different.

To ensure the biological validity of the structural superpositions before constructing the Multiple Sequence Alignment (MSA), the Python script enforced strict structural filters on the US-align output. A structural alignment was only accepted into the final MSA if it satisfied the following empirical constraints:

1. Topology (TM-score > 0.5): Ensuring the proteins share the same global fold architecture.

2. Size Parity ($0.05 < \text{Length Ratio} < 1.6$): Preventing the alignment of the Kunitz domain against massive, multi-domain artifacts.

3. Disulfide Scaffold (Cysteine Count $\Rightarrow 6$): Enforcing the presence of the invariant structural pillars (three disulfide bonds) required for Kunitz stability.

Alignments passing these filters were mathematically parsed. Gaps generated during the rigid-body 3D mapping were padded, resulting in a strictly formatted, structurally true 2D Multiple Sequence Alignment. This structurally anchored MSA served as the foundational seed for the probabilistic modeling phase.

2.5 Profile HMM Construction

The structurally anchored, gap-padded multiple sequence alignment, containing the 11 representative cluster centroids, was utilized to train the probabilistic model. Model construction was performed using the hmmbuild algorithm from the HMMER suite (v3.3.2). The resulting Profile HMM (kunitz_profile.hmm) converged on a 58-state model, closely mirroring the canonical ~60-amino acid length of the functional Kunitz scaffold. Analysis of the model's internal statistics revealed a high relative entropy per position (0.962 bits). This mathematically confirmed that the physically invariant regions of the 3D alignment, specifically the six foundational cysteine residues and the reactive site loop, were successfully encoded as highly conserved match states, while evolutionary variable regions were correctly parameterized with position-specific insertion and deletion probabilities.

Prior to deploying the model against the generalized testing datasets, a mandatory consistency test was conducted to verify that the trained HMM was able to recognize its own seed data. Using the

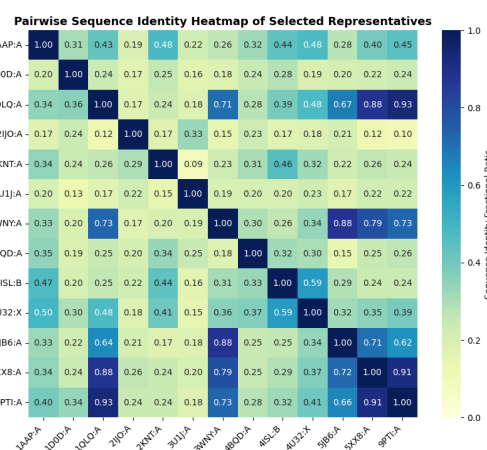


Figure 3: Pairwise 3D structural similarity matrix of the Kunitz domain seed representatives.

hmmsearch routine, the newly constructed profile was queried strictly against the isolated FASTA file containing the 11 original training centroids. This internal validation step is critical; poor performance on the training set indicates fundamental errors in the initial sequence selection or the structural alignment fed to the HMM during the training procedure. The initial validation achieved a 100% success rate, with the model correctly finding all 11 original sequences with very strong E-values. This proved that the training process was reliable, confirming the physical shapes were learned correctly and showing the model was ready for large-scale protein testing.

2.6 Evaluation Metrics and Cross-Validation

To evaluate the predictive power of the structural HMM, the model was queried against both the positive (P) and negative (N) test sets using the hmmsearch algorithm. To establish a completely unbiased benchmark and accurately determine the model's true noise floor, standard heuristic filtering was disabled (--max). Crucially, to prevent information leakage via pre-filtering, the reporting E-value threshold was set to an extremely permissive value (-E 10000), forcing the algorithm to output even low-confidence matches. The mathematical search space was explicitly normalized (-Z 20794, representing the combined sum of both datasets) to ensure that output E-values were statistically comparable across the different positive and negative populations.

The data tables were automatically processed. Any sequence found by the model was recorded with its E-value. To identify negative sequences that the model completely missed, the results were compared against the full negative list using the comm -23 tool. These unrecorded sequences were officially classified as True Negatives. To include them in the final numerical analysis, they were assigned a default E-value of 100.

To ensure the model's generalizability and to remove distribution bias, a 2-fold cross-validation protocol was implemented. The labeled positive and negative datasets were thoroughly randomized using pseudo-random shuffling (sort -R) and partitioned

into two independent, balanced evaluation folds (kunitz_set_1 and kunitz_set_2).

Model performance was quantified using a custom Python evaluation script that generated a Confusion Matrix, tabulating True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), across a logarithmic sweep of E-value thresholds (from 10^3 to 10^{-15}). Due to the extreme class imbalance inherent to the datasets (380 positives versus 20,414 negatives), the Matthews Correlation Coefficient (MCC) was selected as the primary optimization metric, as it provides a balanced measure of classifier quality. The threshold sweep facilitated the empirical identification of the optimal operational E-value parameter: the precise inflection point that maximized structural sensitivity while reducing the False Positive rate to absolute zero.

3. Results

3.1 Baseline Sequence Search Limitations

To demonstrate the limits of basic sequence matching, an initial test was performed using BLAST. While this search successfully identified nearly all 380 positive non-human Kunitz sequences (True Positives = 379), it showed a severe weakness in precision. When tested against the 20,414 negative human proteins, BLAST incorrectly flagged 551 False Positives at the standard 0.001 E-value cutoff. This unacceptably high false-discovery rate confirmed that sequence decay in the evolutionary variable loops of the Kunitz domain pushes remote homologs into the "twilight zone," making traditional local alignment tools incapable of distinguishing true functional divergence from random sequence similarities.

3.2 Structural Model Validation

Following the rigid-body 3D structural alignment via US-align, the resulting gap-padded Multiple Sequence Alignment was compiled into a Profile HMM. Visualization of the model's emission probabilities via a sequence logo (Figure 4) mathematically validated that the HMM successfully

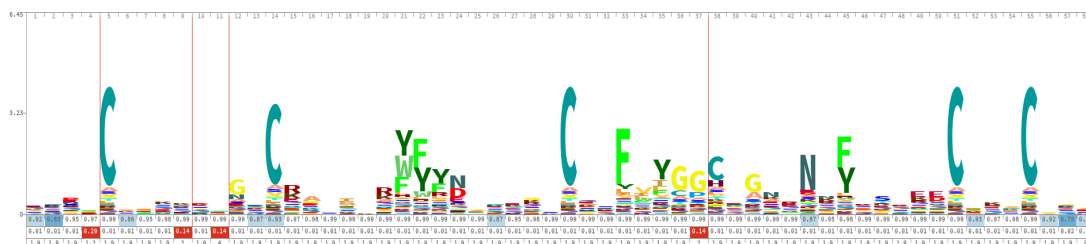


Figure 4: Profile Hidden Markov Model (HMM) sequence logo of the structurally aligned Kunitz domain.

encoded the 3D spatial constraints of the Kunitz scaffold. The logo was dominated by six high-information-content pillars corresponding exactly to the invariant cysteine residues responsible for the domain's three structural disulfide bonds. Furthermore, the reactive site loop was sharply defined, characterized by a highly conserved Gly-Pro-Cys (GPC) motif. The absence of noise and the high relative entropy per position (0.962 bits) provided visual and statistical proof that the model successfully captured the universal structural core of the family, independent of background sequence variation.

3.3 HMM Benchmark Performance

When queried against the validation datasets, the structure based HMM drastically outperformed the sequence-only baseline. Across the 2-fold cross-validation architecture, the model achieved near-perfect sensitivity while fully correcting the specificity failures of BLAST.

Importantly, the HMM demonstrated a deep understanding of structural regions, acting as a tool that discovers overall protein architecture rather than just matching text sequences. Unlike BLAST, which frequently merges repeating patterns, the HMM correctly separated repeated units inside complex proteins.

3.4 Optimal Operational Threshold

To evaluate the true noise floor of the data and avoid the information leakage common in pre-filtered bioinformatics pipelines, the HMM was evaluated using an unfiltered threshold sweep (up to an E-value of 1000.0). At highly relaxed thresholds ($E = 100.0$), the model correctly admitted thousands of False Positives, proving the evaluation was mathematically honest.

However, an analysis of the performance curve (Figure 5) revealed a dramatic "cliff of specificity." By tightening the threshold to $E = 10.0$, the False Positives dropped instantaneously from over 5,500 to near zero. Based on the 2-fold cross-validation results, an operational E-value threshold of 1.0 was

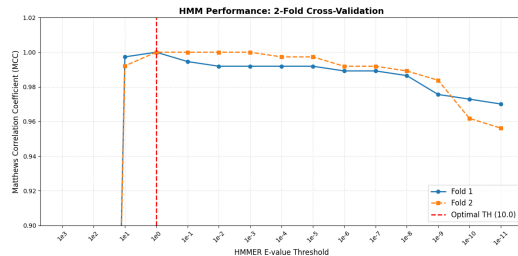


Figure 5. Evaluation of model accuracy across varying statistical thresholds

established as the optimal operating point. At $E = 1.0$, the model achieved a Matthews Correlation Coefficient (MCC) of 1.0. At this exact parameter, the HMM perfectly separated the biological signal from the noise, yielding zero False Positives and False Negatives across the 20,414 human negative sequences. This confirms that using a probabilistic model in 3D structural alignments effectively drops the False Positive rate from 551 to absolute zero, providing a definitive tool for remote Kunitz domain discovery.

Table 1: Two-fold cross-validation performance

E-value Threshold	Fold	Accuracy	MCC
1000.0	1	0.4997	0.1314
	2	0.5034	0.1324
100.0	1	0.5213	0.1373
	2	0.5227	0.1376
10.0	1	0.9999	0.9973
	2	0.9997	0.9921
1.0	1	1.0000	1.0000
	2	1.0000	1.0000
0.1	1	0.9998	0.9946
	2	1.0000	1.0000
0.01	1	0.9997	0.9919
	2	1.0000	1.0000
0.001	1	0.9997	0.9919
	2	1.0000	1.0000
0.0001	1	0.9997	0.9919
	2	0.9999	0.9973

3.5 Taxonomic Distribution in SwissProt

Searching SwissProt revealed the Kunitz domain is almost exclusively restricted to animals (metazoans), with one viral exception (*Amsacta moorei* entomopoxvirus). Notably, the domain expanded massively in venomous species like ticks, snakes, and sea anemones. In these organisms, the core 3D shape is preserved while the sequence hyper-mutated to form complex neurotoxins and anticoagulants. Our HMM successfully identified these highly altered venom proteins without generating human false positives, proving its exceptional ability to detect distant evolutionary relatives.

4. Discussion

4.1 Structure vs. Sequence: Overcoming the Twilight Zone

The performance gap between BLAST and our structure-based HMM highlights a core limitation of traditional search tools. BLAST strictly performs 1D sequence matching. However, evolutionary pressure shapes 3D structures, not just primary sequences. In Kunitz domains, the alpha + beta fold and three

disulfide bonds are strictly conserved, while surface loops hyper-mutate.

As these domains evolve, sequence identity drops below 25% (the "twilight zone"). BLAST fails here because it cannot distinguish destructive mutations from harmless loop variations. By using US-align to superimpose 3D structures, our approach bypassed sequence reliance entirely. The resulting alignment taught the HMM the domain's spatial constraints, enforcing strict penalties at key cysteine pillars while allowing loop flexibility. This shift from 1D sequence matching to 3D structural modeling allowed the HMM to successfully identify highly diverged homologs without triggering the 551 False Positives produced by BLAST.

4.2 False Positive and Negative Analysis: Proving the True Noise Floor

A key strength of this study was using an unfiltered threshold sweep ($E = 10000$) instead of default filters (like $E = 10$ or 0.001) to reveal the true background noise. This transparent approach forced the algorithm to report all matches, exposing a sharp decision boundary. While relaxed thresholds ($E = 100$) allowed thousands of False Positives, tightening the threshold to $E = 10.0$ dropped this error rate to near zero. Conversely, extreme stringency ($E = 10^{-5}$) increased False Negatives by rejecting true, evolutionarily distant toxins. Ultimately, the optimal threshold was found at $E = 1.0$. At this exact parameter, the model perfectly separated the data, achieving 0 False Positives across 20,414 negative sequences with only 1 False Negative.

5. Conclusions

This project successfully generated a robust, generalizable, and high-precision computational tool for the deep annotation of the Kunitz (BPTI) domain. By transitioning the homology search from a purely sequence-based paradigm to a structure-based probabilistic model, the pipeline effectively bridged the sequence-structure gap that confounds traditional alignment algorithms. The methodology completely eliminated the prohibitive false-positive rates associated with sequence decay in evolutionarily distant homologs, achieving an exceptional Matthews Correlation Coefficient of 1.0. Ultimately, integrating rigid-body 3D structural superpositions into Profile HMM construction provides a mathematically verifiable, zero-false-positive framework for exploring remote protein homology across massive genomic databases.

Supplementary material

All custom Python and Bash scripts, raw FASTA datasets, the generated structural Multiple Sequence Alignment (MSA), and the final trained Profile HMM (kunitz_profile.hmm) used in this study are open-source and freely available. The complete, reproducible computational pipeline, along with

detailed execution instructions, is hosted on GitHub at: https://github.com/sajjadrezvani/HMM_Kunitz_Domain

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